

PARTS LIST

Semiconductors:

IC1	- LM317 positive adjustable voltage regulator
IC2	- LM337 negative adjustable voltage regulator
D1-D4	- 1N4007 rectifier diode
D5-D8	- 1N5408 rectifier diode
LED1-LED4	- 5mm LED
ZD1, ZD2	- 1.2V, 1.3W Zener diode or reference IC

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1, R2	- 5.6-kilo-ohm
R3, R4	- 750-ohm
R5, R6	- 240-ohm
R7, R8	- 3.3-kilo-ohm
R9	- 100-ohm, 2W
VR1, VR2	- 1.5-kilo-ohm potentiometer

Capacitors:

C1, C2	- 1000 μ F, 40V electrolytic
C3, C4, C9-C12, C20, C21	- 0.33 μ F ceramic disc
C5, C6	- 47 μ F, 25V electrolytic
C7, C8	- 100 μ F, 25V electrolytic
C13	- 0.1 μ F, 630V polyester
C14-C17	- 22nF ceramic disc
C18-C19	- 2200 μ F, 40V electrolytic

Miscellaneous:

CON1-CON4	- 3-pin connector
F1, F2	- 1A fuse
J1, J2	- 3-pin jumper connector
X1	- 230V AC primary to 15V-0-15V, 1A secondary transformer

depend on the selected ICs and the required output voltage range. The minimal load current for the IC1 and IC2 should be respected in all cases.

Example 1. If you use LM317 as IC1, you can select $R5 = 240\text{-ohm}$, $VR1 = 1.5\text{-kilo-ohm}$ and ZD1 reference diode for 1.2V. You will have regulation of the output voltage V_3 (at point 1 of CON3) from ground up to around 8V. Fortunately, in most cases you can neglect the current of adjustment pin of LM317/LM337. In that case you will have:

$$V_{out}(\text{positive}) = V(\text{LM317}) \times (1 + VR1/R5) + V_{ref1}$$

$$V_{out}(\text{negative}) = V(\text{LM337}) \times (1 + VR2/R6) + V_{ref2}$$

$$V(\text{LM317}) = V(\text{LM337}) = 1.25V$$

V_{ref1} is from ZD1 and V_{ref2} is from ZD2. ZD1 and ZD2 can be the low-cost 1.3W Zener diodes with appropriate voltages. Also, instead of ZD1 and ZD2 you may use an IC (such as LM336 or TL431) for the

reference voltages 1.2V, 2.5V, 5V, etc.

Jumpers J1 and J2 give the options to select positive and negative offsets for IC1 and IC2. In each case you should put the Zener diodes ZD1 and ZD2 in the desired direction. Offset outputs can be obtained at CON2.

Example 2. If you use 7805 as IC1, you can select $R5 = 510\text{-ohm}$, $VR1 = 510\text{-ohm}$ and ZD1 reference diode for 5.1V (such as 1N4733). You will have regulation of the output voltage V_3 from around ground level to above 5V. So now you will have:

$$V_3 = V_{out}(\text{positive}) = V(78xx) \times (1 + VR1/R5) + I_d \times VR1 + V_{ref1}$$

$$V_4 = V_{out}(\text{negative}) = V(79xx) \times (1 + VR2/R6) + I_d \times VR2 + V_{ref2}$$

Here $V(78xx)$ and $V(79xx)$ are output voltages of the fixed regulator (xx denotes the number of the series, for example 7805 for 5V regulator). I_d is the current (quiescent) from the ground pin (bias current) of the regulator 78xx or 79xx.

Preferably, you should have:

$$I_d(R5, VR1) \leq 5 \times I(\text{IC1})$$

$$I_d(R6, VR2) \leq 5 \times I(\text{IC2})$$

But you cannot expect that in most of the cases due to lot of power loss in $R5$, $VR1$, $VR2$ and $R6$.

Input power supply. Fig. 2

shows the power section of the regulator with transformer and rectifiers. For a modular approach you should

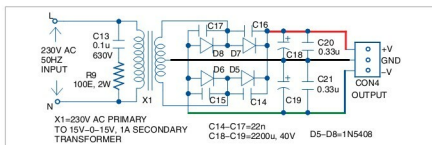


Fig. 2: Input power supply

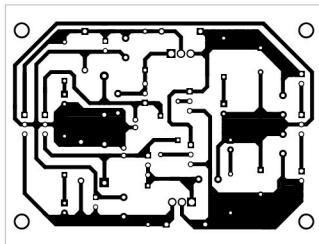


Fig. 3: An actual-size PCB pattern for the universal power supply

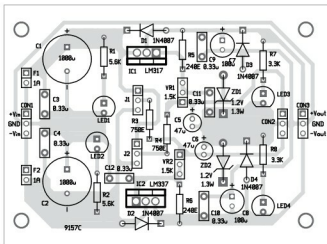


Fig. 4: Component layout for the PCB in Fig. 3